Worksheet 8.1

# Speed and velocity

Student: ..... Class: .....

## Calculating average speed

**1.** The average speed ($v$) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

where $d$ = distance travelled and $t$ = time taken.

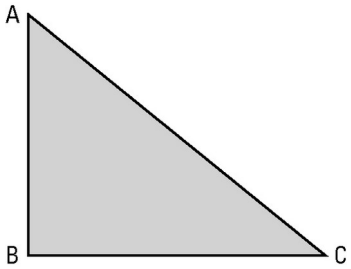
Complete the following table. Make sure you include the appropriate units for each quantity.

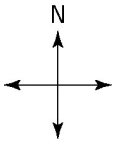
| Distance travelled | Time taken | Average speed |
|---|---|---|
| 100 m | 20 s | |
| 3 km | | 60 km/h |
| | 15 s | 20 m/s |
| 2 km | | 25 m/s |

## Average speed and average velocity

**2.** Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.

AB = 600 m
BC = 800 m




Both vehicles take 70 s for the trip.

Perform the necessary calculations to complete the following table.

| Average speed of X | Average velocity of X | Average speed of Y | Average velocity of Y |
|---|---|---|---|
| | | | |

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